Machine Condition Classification

Model development, validation, and Kaggle submission

Team Name : Beyond Dimension
Team Members : Rajat Aswani, Rishabh Maurya, Sanjiv Mishra
Github Link : https://github.com/rajataswani/mlAcadsBeyondDimension
Competition: Industrial Machine Condition Classification
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Executive Summary

This project predicts one of three machine conditions from sensor measurements: normal operation, thermal failure, or mechanical failure. The supplied labeled dataset contains 16,000 rows, and the unlabeled test set contains 4,000 rows. The work compared a simple baseline, linear and tree-based models, two boosting methods, feature engineering, and a probability blend.

The best validation result was a 50/50 blend of CatBoost and tuned HistGradientBoosting, with 88.68% out-of-fold accuracy in shuffled, stratified five-fold validation. A later submission moved the reported leaderboard position to 14th. Validation and public leaderboard results are estimates from different data splits; neither reveals the final private-test score.

1. Problem and Objective

The task is supervised multiclass classification. Each row describes a machine using operational and sensor features, and the target condition is coded 0, 1, or 2. The goal is to predict the correct condition for unseen machines. The competition metric is accuracy, the fraction of test examples assigned the correct class.

The three labels are 0 for normal, 1 for thermal failure, and 2 for mechanical failure. Because the synthetic dataset is designed to resemble predictive-maintenance data, model performance should be interpreted as performance on this competition dataset rather than as a validated industrial safety system.

2. Data and Quality Checks

The training data has 16,000 rows and 13 columns, including the target. The test data has 4,000 rows and 12 feature columns. The sample submission contains 4,000 machine IDs and the required condition column. The unique unit_id is an identifier, so it was excluded from the predictive features. The audit found 493 missing feature values in training and 123 in test, with no missing target or IDs, duplicate IDs, or train/test ID overlap.

Check	Training data	Test data
Rows	16,000	4,000
Target column	condition	Not provided
ID column	unit_id	unit_id
Missing values	Present in sensor fields and unit_type	Present in sensor fields and unit_type
Target counts	0: 8,000; 1: 4,000; 2: 4,000	Unknown

The target distribution is 50% normal, 25% thermal failure, and 25% mechanical failure. A model that always predicts the majority class would therefore score 50% accuracy, but would fail to identify either failure class. Balanced accuracy was tracked alongside accuracy during development to reveal performance across all three classes.

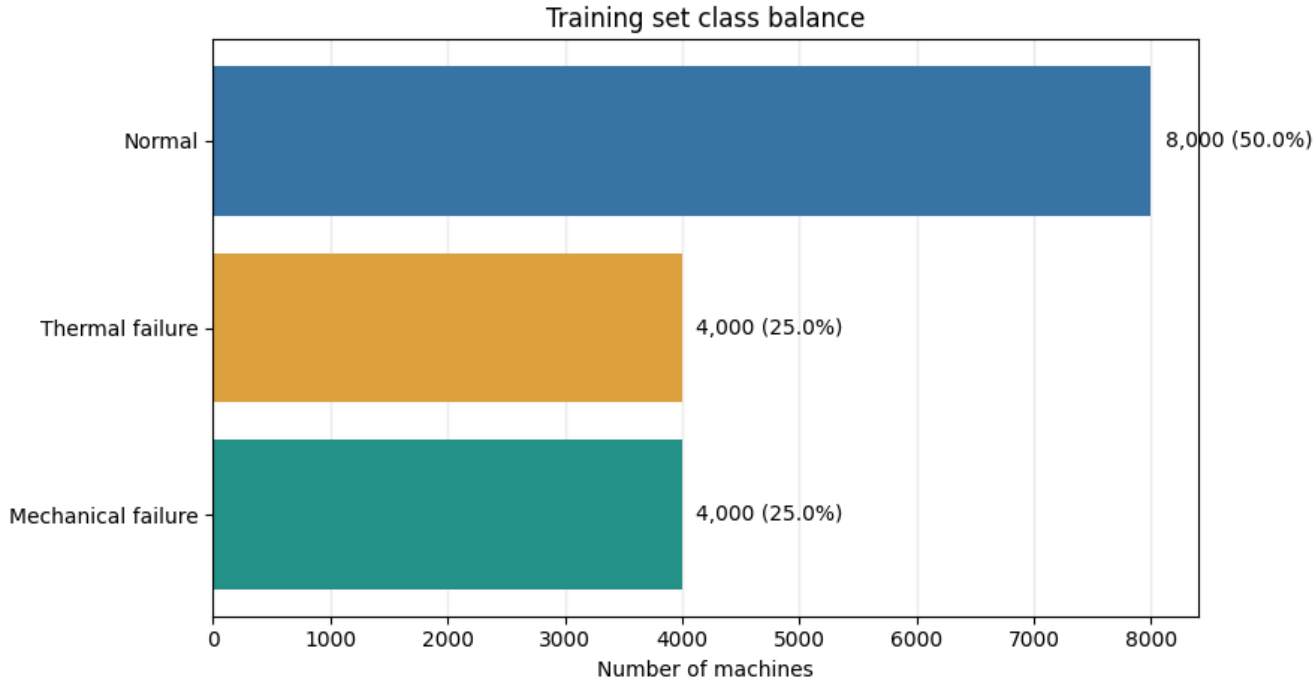


Figure 1. Class counts in the 16,000 labeled training rows.

3. Preprocessing and Feature Construction

The numeric sensor fields contain missing values, and unit_type also has missing entries. In the initial scikit-learn pipelines, numeric values were filled with each fold's training median, and categorical values were filled with the most frequent training category. This keeps validation data from influencing the imputation values. Categorical values were one-hot encoded for the linear and HistGradientBoosting pipelines. Numeric scaling was used for logistic regression; tree-based methods do not require it.

CatBoost was supplied with unit_type explicitly marked as categorical and its missing category values represented as “Missing.” The unique unit_id was omitted throughout the selected model comparisons because it identifies a row rather than measuring machine condition.

Two domain-motivated features were tested: process temperature minus ambient temperature, and the product of drive torque and shaft speed as a rough mechanical power proxy. Engineered CatBoost reached 88.54% mean five-fold accuracy, which was lower than the 88.68% score of the probability blend. The engineered model was therefore not selected as the final candidate.

4. Validation Design and Metrics

Models were compared with shuffled five-fold stratified cross-validation using random_state=42. Stratification keeps the class proportions similar in each fold. Each model is trained on four folds and evaluated on the remaining fold; the process repeats until every labeled row has been held out once. Pipelines perform preprocessing within the training fold.

Accuracy is the competition metric. Balanced accuracy is the average recall across the three classes, so each condition contributes equally even though normal machines are more common. Fold-to-fold standard deviation summarizes how much a model’s accuracy varied across the five validation folds. It is not a confidence interval for private-test performance.

5. Models Compared

Model	Mean CV accuracy	Balanced accuracy	Notes
Most common class	50.00%	33.33%	Baseline
Logistic regression	82.54% ± 0.31 pp	79.88%	Linear baseline
Random forest	87.92% ± 0.36 pp	85.79%	Bagged trees
Extra Trees	87.07% ± 0.13 pp	84.22%	Randomized trees
HistGradientBoosting	88.36% ± 0.32 pp	86.50%	Initial boosting model
Tuned HistGradientBoosting	88.41%	Not recorded	Best search result
CatBoost	88.49% ± 0.15 pp	86.64%	Categorical boosting
Engineered CatBoost	88.54% ± 0.30 pp	Not recorded	Two added features
50/50 CatBoost + HistGB	88.68%	Not recorded	Out-of-fold probability blend

The blend’s 88.68% score was calculated from out-of-fold probabilities generated on the same five-fold scheme. Blend weights were compared on these out-of-fold predictions, so this score is useful for model selection but may be optimistic. The blend improved over either individual model in that comparison; the small differences between the individual boosting models should not be treated as proof that one will always generalize better.

The selected HistGradientBoosting settings were 100 boosting iterations, learning rate 0.05, 31 maximum leaves, 30 minimum samples per leaf, and zero L2 regularization. CatBoost used 800 iterations, depth 7, and learning rate 0.05. These settings document the reported experiments; they were not exhaustively optimized.

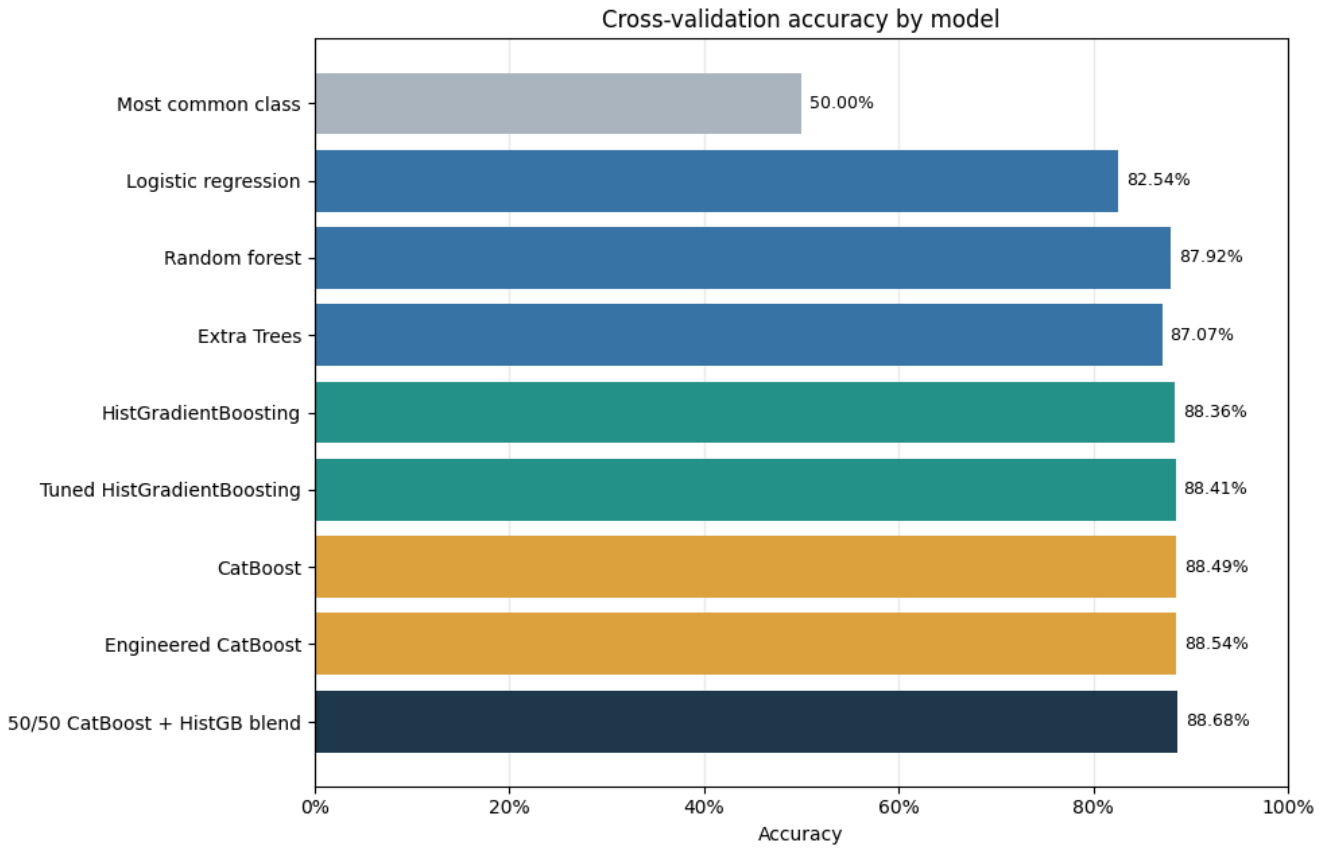


Figure 2. Mean five-fold accuracy for the tested models and the selected 50/50 blend. The blend's value is the pooled out-of-fold accuracy.

6. Error Analysis

The confusion matrix below comes from out-of-fold predictions for the initial HistGradientBoosting model, before its hyperparameter search. Overall accuracy was 88.36%. The model recognized normal machines most reliably, while failure examples were more often confused with another condition.

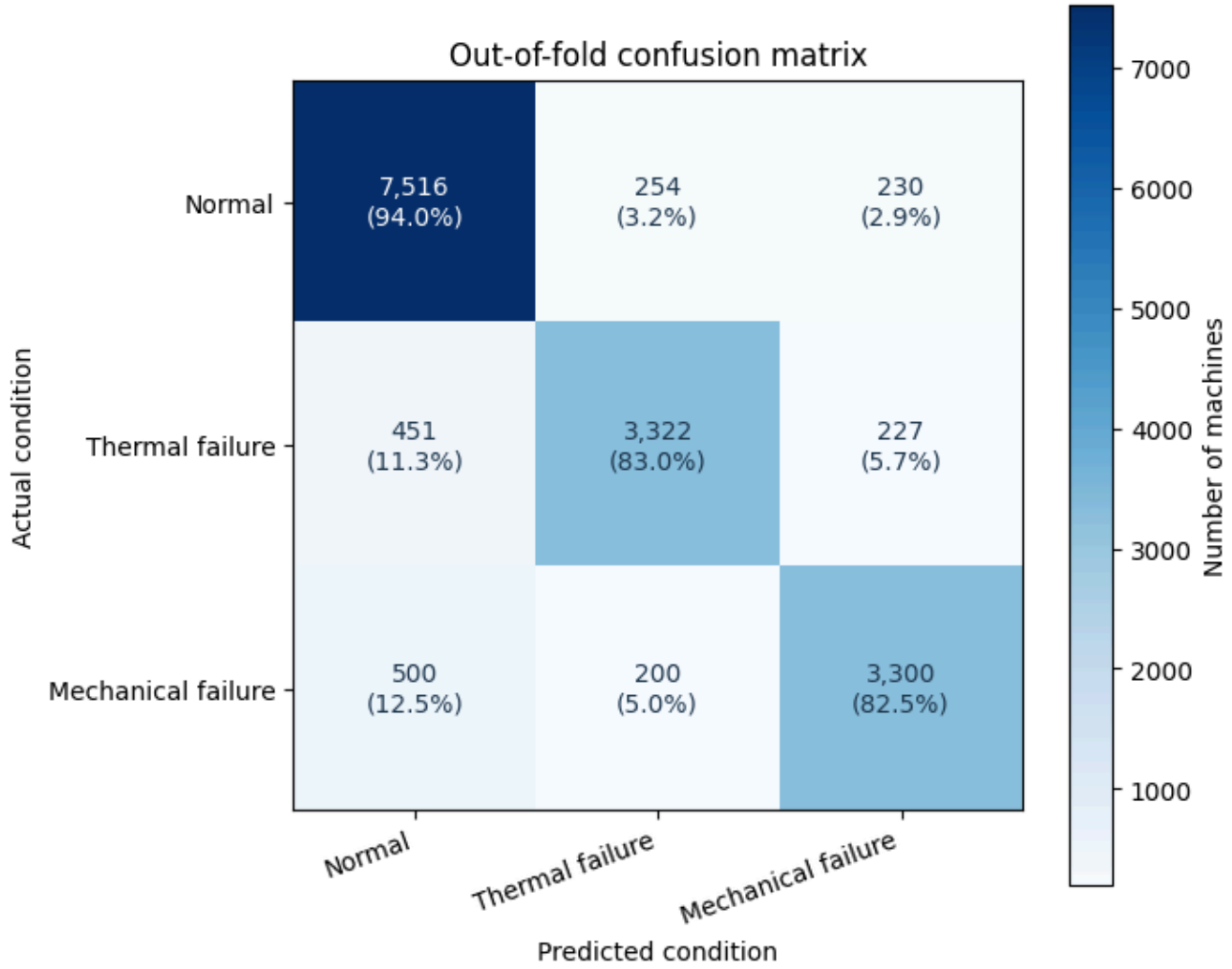


Figure 3. Out-of-fold confusion matrix for initial HistGradientBoosting. Diagonal cells are correct predictions.

Actual condition	Precision	Recall	F1 score	Rows
Normal	88.77%	93.95%	91.29%	8,000
Thermal failure	87.98%	83.05%	85.44%	4,000
Mechanical failure	87.84%	82.50%	85.08%	4,000

Of 4,000 thermal-failure examples, 3,322 were correctly classified; 451 were predicted as normal and 227 as mechanical failure. Of 4,000 mechanical-failure examples, 3,300 were correct; 500 were predicted as normal and 200 as thermal failure. This pattern suggests that separating the two failure types is harder than identifying normal operation.

7. Selected Candidate and Kaggle Result

The selected candidate averages the class probabilities from CatBoost and tuned HistGradientBoosting equally, then chooses the class with the largest average probability. Both models are fitted on all 16,000 labeled rows before predicting the 4,000 test rows. The generated submission uses the sample template's unit_id order and the required columns unit_id and condition, with predictions restricted to classes 0, 1, and 2.

The first submitted file, submission_machine_condition.csv, was accepted with a public score of 0.88312 and an initial reported position of 19th out of 20. After a later blended submission, the reported position moved to 14th out of 20. The exact public score for that later entry was not provided, so no score is assigned to it here. Leaderboard position is relative to the other participants and does not determine the private-test result.

8. Limitations

- The final private-test accuracy is unknown until the competition publishes it.
- The 50/50 blend weight was chosen after comparing several weights on the out-of-fold predictions; the reported blend score is therefore a selection estimate rather than a fully untouched evaluation.
- The confusion matrix describes the initial HistGradientBoosting model, not the final blend. Per-class performance for the final blend was not separately reported.
- The dataset is synthetic. Results may not transfer to real industrial equipment, where sensors, failure rates, operating conditions, and costs can differ.
- Accuracy treats all errors equally. In a real maintenance setting, missing a failure may be more costly than a false alarm, so recall, calibration, and cost-sensitive evaluation would also be needed.

9. Conclusion

The project established a complete classification workflow: data checks, missing-value handling, a variety of model families, stratified cross-validation, error analysis, feature experiments, and a correctly structured Kaggle submission. The best tested validation candidate was the 50/50 CatBoost–HistGradientBoosting probability blend at 88.68% out-of-fold accuracy. The public leaderboard position later improved to 14th of 20. Because validation and leaderboard results are estimates, the project's main conclusion is that the blended pipeline performed consistently on the available evaluation data; its private-test performance remains unknown.